In my work what is T2 about

In your quantum work, T_2 is the coherence time: how long a qubit retains the phase relationship that makes quantum interference possible. It measures the lifetime of that coherence—not intelligence, consciousness, or the lifetime of the notebook itself.[1] [2]

What is fading?

A qubit can retain its 0/1 measurement probabilities while losing the phase information connecting those possibilities; T_2 describes the decay of the off-diagonal elements of its density matrix.[2]

For a simple exponential model, the coherence magnitude follows

$$C(t) = C(0)e^{-t/T_2}.$$

At $t = T_2$, approximately **37% of the initial coherence remains**—it is not an abrupt expiration time.[2]

Three related quantities

Quantity	What it describes
T_1	Energy relaxation: how quickly an excited qubit relaxes toward equilibrium. [3] [4]
T_2	Phase-coherence decay, including contributions from energy relaxation and pure dephasing; commonly characterized with an echo experiment. [1]
T_2^*	Ramsey coherence decay, which also includes effects of slow frequency variation or inhomogeneous broadening. [5]

The measurement protocol matters: a Ramsey-derived T_2^* and an echo-derived T_2 are not interchangeable.[5] [1]

Why it matters for you

For your mixing and recovery experiments, distinguish **how much coherence you start with** from **how quickly it decays**: in a fit such as $Ae^{-t/T_2^*}\cos(2\pi ft + \phi) + B$, amplitude A and decay time T_2^* are separate parameters. A smaller signal alone does not demonstrate a shorter coherence time.[6]

The clean notebook I just built did not measure T_2. It compared recovery fidelity at specified noise probabilities, without a physical delay sweep; its fidelity improvements therefore cannot yet be called improvements in T_2.

To test that next, prepare coherence, vary the storage delay, measure the interference signal, and fit its decay—with and without recovery under matched conditions. Ramsey and Hahn-echo experiments provide established starting protocols. [1] [5]

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1. https://qiskit-community.github.io/qiskit-experiments/stubs/qiskit_experiments.library.characterization.T2Hahn.html
2. https://qiskit-community.github.io/qiskit-experiments/manuals/characterization/t2hahn
3. https://qiskit-community.github.io/qiskit-experiments/stable/0.13/apidocs/library.html
4. https://qiskit.github.io/qiskit-aer/stubs/qiskit_aer.noise.thermal_relaxation_error.html
5. https://qiskit-community.github.io/qiskit-experiments/_modules/qiskit_experiments/library/characterization/t2ramsey.html
6. https://qiskit-community.github.io/qiskit-experiments/manuals/characterization/t2ramsey.html
7. https://qiskit-community.github.io/qiskit-experiments/dev/manuals/characterization/t2ramsey.html
8. https://qiskit.github.io/qiskit-aer/stubs/qiskit_aer.noise.NoiseModel.html
9. https://qiskit.github.io/qiskit-aer/release_notes.html
10. https://qiskit.github.io/qiskit-aer/tutorials/2_device_noise_simulation.html
11. https://qiskit.github.io/qiskit-aer/apidocs/aer_noise.html
12. https://qiskit.github.io/qiskit-aer/tutorials/3_building_noise_models.html
13. https://qiskit-community.github.io/qiskit-experiments/stable/0.5/_modules/qiskit_experiments/library/characterization/t2hahn.html
14. https://qiskit-community.github.io/qiskit-experiments/stable/0.12/stubs/qiskit_experiments.library.characterization.T2Hahn.html
15. https://qiskit.github.io/qiskit-aer/stubs/qiskit_aer.noise.device.basic_device_gate_errors.html